

Investigation Log: Periodicity of the Octal Game 0.45

(a negative result, and the reading rule it produced)

Aryan Padarthy

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Abstract

After four consecutive candidate results in low-dimensional topology turned out to be already known (`manuscript/log.tex`, Log Entries 17, 19, 23), this log records a deliberate pivot to combinatorial game theory and its outcome: a fifth miss, and the sharpest one so far. The octal game 0.45 was selected as a target on the belief that its conjectured period-20 nim-sequence was unproved. It is not. A. Flammenkamp’s octal-game database lists .45 in its table of *Nontrivial Octal Games with known Structure* with a “solved” date of **1956** — that is, the game was settled by Guy and Smith in the paper that introduced the notation. Independently of that citation, the Guy–Smith periodicity theorem closes the game in half a page and requires the nim-sequence to be checked only as far as $n = 2p + 2\ell + t = 1038$; that argument is re-derived from scratch and verified here. What survives is a fully validated Sprague–Grundy engine, which reproduces all four of Flammenkamp’s tabulated statistics for .45 exactly, and a reading rule stated in Entry 5: the failure this time was not inferring openness from silence, but misreading a primary source that gave the answer in a column I did not parse. Both the wrong claim and its correction are left in place below, in order.

Entry 1: Why This Problem

Date: September 23, 2026

After the knot-theory audit’s fourth consecutive miss, a different area was sought, following the standard applied throughout: find a specific, well-defined, checkably-open target before committing effort, rather than assume openness.

The area. Octal games (Guy–Smith notation $0.d_1d_2d_3\dots$) are take-and-break Nim variants; whether every finite one has an eventually periodic nim-sequence is a genuine open conjecture in the field, and Flammenkamp’s page does record 65 unsettled 2-place and 8 unsettled 3-place octal games. That much was right.

The candidate. 0.45: remove 1 token from a heap only by splitting the remainder into two nonempty heaps (needs heap ≥ 3); remove 2 tokens either to end the heap (heap = 2) or by splitting the remainder (needs heap ≥ 4). Its database entry gives preperiod 498, period 20 — by a wide margin the smallest such pair among the small-code games examined (.356: 7315/142; .156: 3479/349; .644: 3256/442). 0.45 is not among the four games (.127, .16, .376, .56) proved periodic by Gangolli–Plambeck, and I read its absence from that list, plus the field’s “only about twenty proved periodic” framing, as evidence that it was unproved. *This inference was wrong; see Entry 5.*

Entry 2: A Validated Engine

Date: September 23, 2026 **Command:** `python src/octal_games/grundy.py src/octal_games/grundy.py` computes Sprague–Grundy sequences for arbitrary octal codes from scratch (a direct mex computation, no special-casing). This part of the work stands, and is the most reusable thing the investigation produced.

- **Kayles** (0.77): the engine’s own search finds preperiod 71, period 12 — an exact match to R. K. Guy’s 1949 hand-proved result.
- **Four period cross-checks:** 0.45 (498, 20), 0.356 (7315, 142), 0.156 (3479, 349), 0.644 (3256, 442), all reproduced exactly.
- **Four statistic cross-checks on .45**, added during the audit and stronger than the period check alone, since they test four independent columns of the same database row rather than one:

Flammenkamp column	his value	computed	
rare (indices with even popcount G)	11	11	match
miss ($\#\{n : G(n) \neq G(n - 20)\}$)	67	67	match
last+t (largest rare index + t)	200	200	match
maxG	8	8	match

A second, independent engine (vectorised, different inner loop) was written during the audit and agrees with the first on every value of G for $0 \leq n \leq 3000$; it extends the computation to $N = 200,000$, where the period-20 relation still holds with zero exceptions past $n = 517$.

Entry 3: Three Structural Findings

Date: September 23, 2026

Computed to $N = 60,000$, later extended to $N = 200,000$.

1. **Boundedness.** $G(n) \in S = \{1, 2, 4, 7, 8\}$ for all $199 \leq n \leq 200,000$, with zero exceptions; below 199 there are exactly 11 exceptions, namely $n = 0, 1, 6, 11, 19, 31, 78, 88, 98, 106, 198$.
2. **Stabilization.** The mex-determining reachable set at n draws in principle on all splits $a + b = n - 1$ or $n - 2$, a range growing without bound. Restricted to the values $\{0, \dots, 8\}$ that can affect the mex, it is determined by the first A split terms with $A \leq 98$ across 416 sampled $n \in [500, 60,000]$.
3. **Closure.** Combining 1 and 2, the mex-relevant reachable set, and hence $G(n)$, depends only on $n \bmod 20$ once the relevant window lies past $n = 199$.

Finding 2’s bound is worth one remark, because it is the one place where my own computation and the database met independently: the widest offset the recursion actually needs is $198 + t = 200$, and 200 is exactly the value in Flammenkamp’s `last+t` column. I computed that number without knowing what the column meant.

Entry 4: The Correct Argument

Date: September 25, 2026

Command: `python verification/prove_octal_45_periodicity.py`

An earlier version of this log built a bespoke proof on Entry 3: a bounded value set S , the ten-index exceptional set $R = \{1, 6, 11, 19, 31, 78, 88, 98, 106, 198\}$, an XOR-closure lemma ($x, y \in S$ and $x \oplus y \leq 8$ imply $x \oplus y \in C = \{0, 3, 5, 6\}$), and a reachable-set formula proved by strong induction. That argument had one acknowledged gap: that the bulk contribution realises *all* of C and not merely a subset of it, which I could not establish without assuming the very periodicity being proved. The whole apparatus is unnecessary. None of S , C , R or boundedness is needed.

Theorem 1 (Guy-Smith, specialised to 0.45). *Let G be the nim-sequence of 0.45, let $\ell = 20$, $p = 498$, and let $t = 2$ be the index of the last nonzero code digit. If $G(m) = G(m - \ell)$ for every m with $p + \ell \leq m \leq 2p + 2\ell + t$, then $G(n) = G(n - \ell)$ for all $n \geq p + \ell$.*

Proof. For $n \geq 4$ the game's moves give $\text{Reach}(n) = \{G(a) \oplus G(b) : a + b = n - i, a, b \geq 1, i \in \{1, 2\}\}$ and $G(n) = \text{mex} \text{Reach}(n)$; there is no move leaving exactly one heap, and the move that empties the heap applies only at $n = 2$. Induct on $n > 2p + 2\ell + t$, assuming $G(m) = G(m - \ell)$ for all $p + \ell \leq m < n$.

$\text{Reach}(n) \subseteq \text{Reach}(n - \ell)$. Take $a + b = n - i$ with $a \leq b$, $a, b \geq 1$. Then $b \geq (n - i)/2 \geq (n - t)/2 \geq p + \ell$ and $b \leq n - i - 1 < n$, so the hypothesis gives $G(b) = G(b - \ell)$, and $b - \ell \geq p \geq 1$. Hence $\{a, b - \ell\}$ is a pair of nonempty heaps summing to $(n - \ell) - i$ with the same XOR. Heaps are unordered, so no condition relating a and $b - \ell$ is required — this is the step my earlier write-up split into two cases unnecessarily.

$\text{Reach}(n - \ell) \subseteq \text{Reach}(n)$. Take $a + b = (n - \ell) - i$ with $a \leq b$. Then $b \geq (n - \ell - i)/2 \geq p$ and $b + \ell \leq n - i - 1 < n$, so $G(b + \ell) = G(b)$ and $\{a, b + \ell\}$ sums to $n - i$ with the same XOR.

Neither n nor $n - \ell$ equals 2 in this range, so the terminal move contributes to both or neither. Equal reachable sets have equal mex, so $G(n) = G(n - \ell)$. $\square$

With $p = 498$, $\ell = 20$, $t = 2$ the finite hypothesis runs only to $2p + 2\ell + t = 1038$, which the engine verifies. The last exception to $G(n) = G(n - 20)$ is at $n = 517$, so 498 is exactly the preperiod. Flammenkamp's page states a sharper bound still — $t + \max\{\text{last}, \text{depth}\} = 2 + \max\{200, 37\} = 202$ successive values suffice. `verification/prove_octal_45_periodicity.py` checks the hypothesis window, both inclusions pair by pair over a band above the bound, the preperiod, and the four engine statistics; full output in `results/octal_games/prove_045_periodicity_output.txt`.

Entry 5: Audit — the Fifth Miss, and What Caused It

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Flammenkamp's page carries two tables. The one I read gives per-game sequences and periods. The one I did not read is headed *Nontrivial Octal-Games with known Structure* and ends in a `solved` column. Its row for .45 is

game	per.	prep.	miss	dens.	ty.	rare	l+t	mxG	dep.	au.	cor.	solved
.45	20	498	67	0.1345	0	11	200	8	37	1	0.4	1956

so the game was solved in 1956, and the four statistics I had reproduced as engine validation (**miss**, **rare**, **last+t**, **maxG**) were sitting in the same row as the date that made the project pointless. The page’s 65 unsettled 2-place games are listed separately; .45 is not among them.

This is the fifth consecutive target that turned out to be known, but the failure mode is new and worse. Misses 1–4 came from inferring openness from absence of evidence — nobody had written the result down where I looked. This one came from reading a primary source that stated the answer explicitly, in a column I did not parse, while extracting four other numbers from the same row. Volume of verification did not help: every computational claim in Entries 2 and 3 is correct, and two of them agree with the database to the digit.

Rule adopted. Before any effort is committed to a target, every column of every table in the primary source that names the target must be identified by name, including columns not being used. A source consulted for one field has not been read. Concretely: if a table has a **solved** column, read it first; if the meaning of a column is unknown, that is a blocking question, not a detail. A corollary applies to the literature gate generally — a positive, dated statement of openness must name the specific game or parameter, not the family.

What survives. The engine (Entry 2), validated eight independent ways. The re-derivation of the Guy–Smith shift argument (Entry 4), which is correct mathematics and is now the thing I would cite. The negative record itself, which is the reason this log exists.

What does not. Any claim of novelty for 0.45; the bespoke *S/C/R* proof of Entry 4’s superseded version; and the framing in the previous abstract, which described the boundedness half as “a complete, gap-free proof” of something already proved in 1956.